Exploring Factors Affecting Hourly Demand and Revenue for Taxi Trips in NYC

Among High-Volume For-Hire Vehicle Drivers

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Github repo with commit

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1 Introduction

In recent years, New York City (NYC) taxi transportation has shifted due to the demand for more convenient transportation options and for app-based bookings to save waiting time. High-Volume For-Hire Vehicles (HVFHV), primarily serviced by Uber and Lyft, have dominate the market share not yellow taxi anymore. NYC Taxi & Limousine Commission (NYCTLC) indicates that, yellow taxis account for only a small part of 24 million monthly total trips, while HVFHV services cover 82.6% of the city's needs, with 84,085 licensed vehicles[1]. Thus, the taxi type chosen to analyze is HVFHV.

This study investigates important elements influencing hourly demand and revenue for taxi rides in NYC using Linear Regression and Random Forest Regressor models. Feature importance from these models will be used to assess the significance of different factors. Hourly demand and revenue are defined as the total number of pick-ups and net incomes (include tips and net of commission, surcharges, or taxes) for all HVFHV drivers within a specific hour. The insights gained aim to assist taxi companies and drivers in optimizing vehicle dispatch to boost revenue.

Data from the second half of 2023 is analyzed to determine potential influences for the remaining months of 2024. The primary dataset used is the **HVFHV Trip Records**[2] from NYCTLC, which include trip details like timestamps, locations, trip distance, and financial data. Additionally, the **Taxi Zone Shapefile**[3] and **Taxi Zone Lookup Table**[4] by NYCTLC offer geographic and zone data, which are essential for visualizing and examining the spatial distribution of HVFHV services. The attributes of interest in these datasets include pick-up date and hour, location, and zone, as they are crucial for understanding hourly demand and revenue dynamics. Time-related features help identify peak hours and dates, while location and zone data reveal geographic patterns in taxi usage.

The following two external datasets support further analysis due to their expected high correlation with the research goals. Firstly, the **Primary Land Use Tax Lot Output (PLUTO)**[5] by NYC Department of City Planning provides detailed information on land use and property characteristics. Ultimately, the the **Local Climatological Data (LCD)**[6] measured at NYC Central Park by National Centers for Environmental Information describes the hourly, daily, and monthly weather conditions by lots of characteristics such as temperature and visibility. Features of interest in these datasets include building class and hourly temperature, since they help explore how environmental and urban factors affect passenger behaviors. For example, high temperatures may reduce demand, while higher demand may occur around commercial or residential buildings.

2 Preprocessing

2.1 Data Wrangling

2.1.1 HVFHV Trip Records

Initially, the shape of this dataset is 117,277,281 records with 24 features. For each effective trip, assume `trip_miles` is greater than one mile, `trip_time` is greater than one minute and less than five hours, `base_passenger_fare` is greater than one United States Dollar ($), and `driver_pay` is greater than zero. Following are the important steps to clean data.

- **Convert Units:** Changed the unit of the `trip_time` column from seconds to minutes for better readability and consistency.
- **Drop Unused Columns:** Dropped the columns `originating_base_num` and `on_scene_datetime` due to large number of NULLs (33,581,394) and irrelevance to the analysis. Now the shape of the dataset is 117,277,281 records with 22 features.
- **Outlier Detection and Removal:** Eliminated ineffective data or the data which is violated to assumption.
  - Removed rows where `request_datetime` started with "2023-06-30" or "2024-01-01" and `dropoff_datetime` started with "2024-01-01" since they are out of research timeline.
  - Filtered out rows where: `PULocationID` and `DOLocationID` were out of the valid range (1 to 263), `trip_miles` was less than or equal to 1 mile, `trip_time` was less than or equal to 1 minute or greater than 300 minutes, `base_passenger_fare` was less than or equal to $1, and `driver_pay` was less than or equal to 0 (since the last three are violation of the assumption).
  - Finally, 14% (16,551,502) of original instances were removed, leaving 100,725,779 records and 20 features for feature engineering.

2.1.2 PLUTO Dataset, Taxi Zone Shapefile, and Taxi Zone Lookup Table

Initially, there are 859,012 of rows and 101 attributes in PLUTO dataset, 263 rows and 7 features in Taxi Zone Shapefile, and 265 rows and 4 attributes in Taxi Zone Lookup Table. Following are the crucial steps to handle these three datasets.

- **Drop Unrelated Columns:** Since most of attributes are irrelevant to the research goals and there are large amount of missing vales, only columns `borough`, `bldgclass`, `latitude`, and `longitude` in PLUTO dataset are left for further analysis.
- **Merge Three Datasets Together:** Taxi Zone Lookup Table and Taxi Zone Shapefile were merged on `LocationID` by the inner join to generate a new GeoDataFrame `zone_gdf`. Then performed spatial join on PLUTO datasset and `zone_gdf` by the "within" predicate. This ensure the building's geometry in PLUTO dataset is entirely within the geometry in `zone_gdf`.

The resulted dataframe has 856,087 rows and 6 features. This provides information on the building class, zone, and borough for each location ID based on their latitude and longitude.

2.1.3 Local Climatological Data

Initially, the shape of this dataset is 11,803 rows and 125 columns. Following are the key points to deal with the inconsistency in Local Climatological Data (LCD).

- **Data Filtering and Selection:** Removed the records from January to June as it was out of the research timeline. Since most of attributes did not match the research purpose, only column `DATE`, `HourlyDryBulbTemperature`, `HourlyPrecipitation`, `HourlyVisibility`, and `HourlyWindSpeed` were selected. Now 6,140 records and 5 columns are left.
- **Data Cleaning:**
  - According to LCD Dataset Documentation[7], "T" in `HourlyPrecipitation` indicates trace amount of precipitation. So replaced "T" with 0 for convenience.
  - Converted `DATE` to a timestamp and extracted two new columns `DateOnly` and `Hour`, which helps merge data on `Hour` column easier.Then dropped unused `DATE` column. So there are 6,140 rows and 6 attributes.
- **Imputation:**
  - Imputed NULL values using the previous effective value (also known as forward fill) within each hour since weather is the time series data.
  - Because some days' data were not recorded, a complete range of dates and hours DataFrame was crossed join with existing data to ensure all possible timestamps were included. Then used forward fill to handle any missing values after merging datasets. This leads to 6,144 records and 6 columns.
- **Data Aggregation:** Aggregated the data on `DateOnly` and `Hour` to compute the average values for each hour of each day. Consequently, the final dataframe has 4,416 records and 6 features.

2.2 Feature Engineering

For HVFHV Trip Records dataset, several feature columns were added which help analyze demand and revenue. Following are the important procedures:

- `day_of_week`, `day_type`, and `pickup_hour` were extracted from `pickup_datetime` to indicate whether it was Monday, Tuesday, etc., whether it was a weekday or weekend, and the hour of pick-up. This helps analyze how hourly demand and revenue vary on different days of the week, between weekdays and weekends, and identifies peak or trough hours during certain periods.
- `request_to_pickup_time` was calculated by subtracting the `request_datetime` from `pickup_datetime` to indicate the waiting time in minutes for passengers after they request a HVFHV trip. This metric was then used to calculate the `utilization_rate` as `trip_time`/(`request_to_pickup_time` + `trip_time`). The utilization rate is a crucial metric for understanding driver efficiency, reflecting the percentage of time that high-volume drivers spend actively transporting passengers according to HVFHV driver income rules[8]. Note that this represents a rough utilization rate since the time spent waiting for a dispatch is not available.
- `total_fare_amount` (calculated by `base_passenger_fare` + `tolls` + `bcf` + `sales_tax` + `congestion_surcharge` + `airport_fee` + `tips`), and `total_revenue` (calculated by `driver_pay` + `tips`) were also added. These features help in understanding the total cost for passengers and the total income for drivers, and also help simplify the modelling process by reducing the number of factors.
- Rows containing zero values were removed during this stage to ensure data quality.

The final HVFHV Trip Records dataset now contains 99,692,683 records and 28 features, which are ready for data aggregation.

2.3 Data Aggregation

The final HVFHV Trip Records dataset were aggregated to analyze the trends of hourly demand and revenue across different time series (hours, days, months) and were standardized by dividing by the total number of hours in a day, total number of days, and total number of months in the research timeline respectively, ensuring comparability across different time periods. Supplementary features such as `day_type` and `PULocationID` were also included. Additionally, these independent sub-datasets were saved for subsequent visualization and aggregation with PLUTO and LCD datasets.

For PLUTO and LCD datasets, daily demand and revenue were grouped by building class and location respectively then were aggregated. Apart from that, hourly demand was merged with hourly weather data. These steps facilitate the analysis of geographical and climatic influences on demand and revenue patterns, enabling a comprehensive visualization of these effects.

Ultimately, the final HVFHV Trip Records, hourly demand, and hourly weather datasets were merged together to create a comprehensive dataset for modelling. Missing information for `PULocationID` 57 was identified in PLUTO dataset, and rows with this location were removed in merged dataset to ensure data integrity. Next, this combined dataset was aggregated by hour and underwent One-Hot Encoding for categorical variables, preparing the data for machine learning models. This process preserved the detail of discrete features and resulted in a dataset with 1,103,947 rows and 577 features.

2.4 Feature Selection and Data Transformation

In this phase, the correlation heatmaps were used to identify and exclude features with low relevance to the target variables `avg_total_revenue` and `avg_hourly_demand`. Features with minimal correlation or redundancy were removed to streamline the model and enhance its efficiency. Specifically, attributes like `avg_shared_request_flag` and `avg_request_to_pickup_minutes` were excluded. Additionally, log transformations were applied to numerical features with skewed distributions, such as `avg_trip_miles` and `avg_total_fare_amount`, to stabilize variance and improve model accuracy. This final dataset, now including 1,103,947 records and 560 variables, was then saved for subsequent modelling in section 4. These steps ensured the dataset was well-prepared, with meaningful features and transformed attributes that would contribute to more robust models and accurate interpretation.

3 Analysis and Geospatial Visualisation

This section will discover the interesting facts or patterns for attributes related to demand and revenue. The major summary is demand and revenue peak during commutes and major events, with concentrations in airports and residential regions.

3.1 Time Series Analysis (Hour and Day)

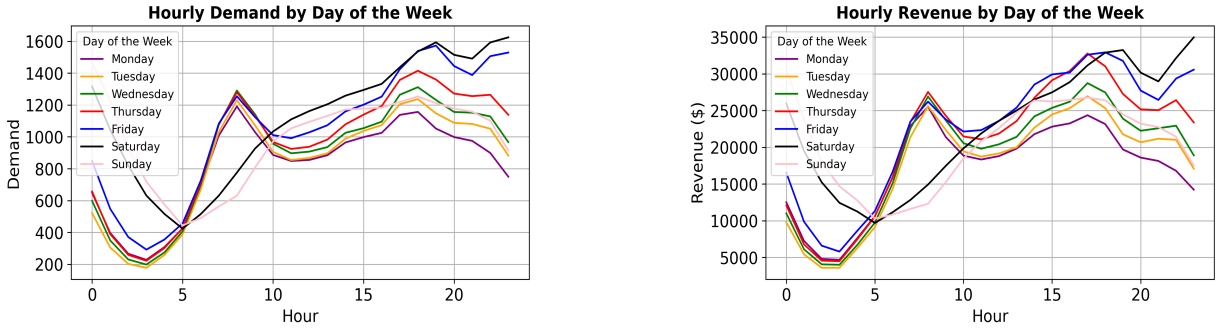


Figure 1: Hourly Pick-up Demand and Revenue by Day of the Week

Figure 1 shows peaks in demand and revenue around 8 AM and 5-7 PM. These peaks likely correspond to morning and evening commutes, when people are traveling to and from work, which increases the demand for taxis. Moreover, these times align with common daily routines and business hours, further driving up both hourly demand and revenue.

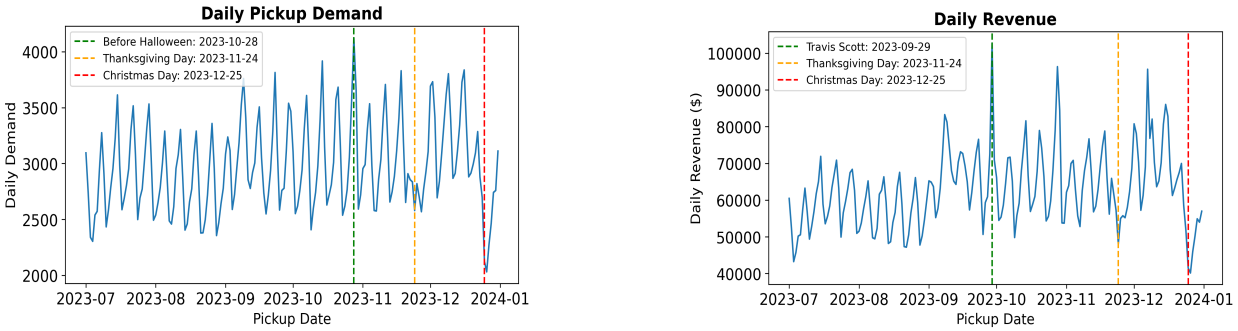
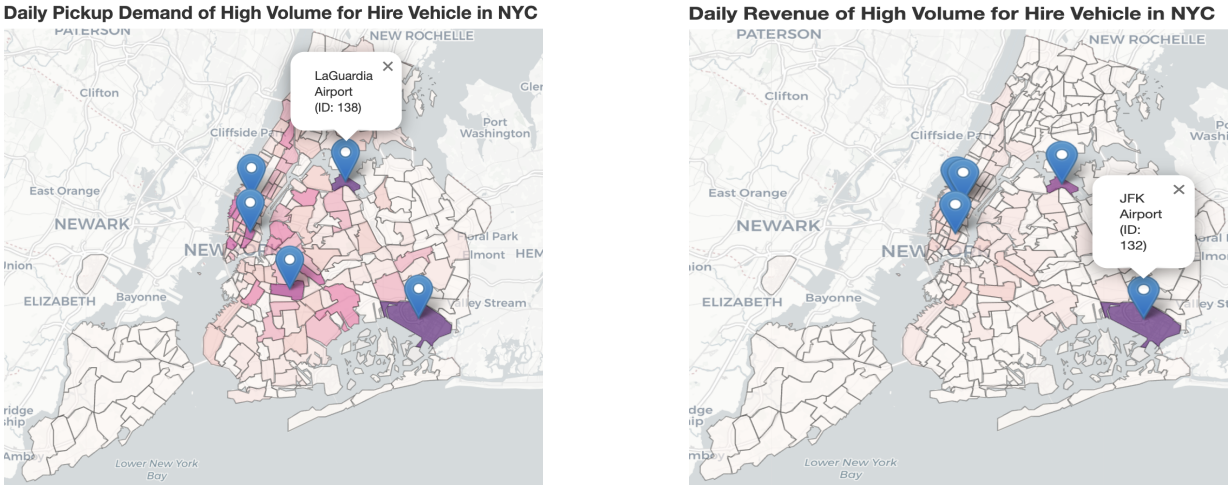


Figure 2: Daily Pick-up Demand and Revenue

Figure 2 highlights similar trends in daily pick-up demand and revenue on specific dates. The highest pick-up demand occurred on October 28, 2023, likely due to pre-Halloween events and celebrations, while the highest revenue was recorded on September 29, 2023, during the Travis Scott concert at Madison Square Garden. Both events, being major entertainment activities, attract large crowds and consequently increase transportation demand and revenue. Conversely, there were notable decreases in demand and revenue on Thanksgiving Day and Christmas Day, with a particularly sharp drop on Christmas. This decline is likely due to reduced transportation needs during these traditional holidays. Furthermore, the observed patterns align with increased demand around major events and vacations, and decreased needs during traditional holiday periods.

3.2 Geographical Analysis (Location and Building Class)



(a) Daily Pick-up Demand

(b) Daily Revenue

Figure 3: Daily Pick-up Demand and Revenue in Different Locations

Figure 3 indicates the highest daily pick-up demand of 10,762 trips occurs at LaGuardia Airport, and the highest daily revenue of $490,265.60 is at JFK Airport. This suggests that airports are major hubs for both high pick-up demand and substantial revenue. In addition, the other top-demanded zones for pick-up demand and revenue are in Manhattan and Brooklyn. Specifically, the Midtown Center, East Village and Times Square/Theatre District in Manhattan and Crown Heights North in Brooklyn are prominent. This distribution revels that these areas are high-density regions with significant

commuting, residential and commercial activities, which likely drive increased transportation needs and revenue. The significance of these locations can be attributed to their popularity as cultural, entertainment, and residential centers, leading to high levels of activity and transportation demand.

Building Class	Daily Demand	Daily Revenue
B	5.17×10^8	$\$9.39 \times 10^9$
A	4.30×10^8	$\$7.84 \times 10^9$
C	3.93×10^8	$\$7.46 \times 10^9$
S	9.47×10^7	$\$1.82 \times 10^9$
D	4.75×10^7	$\$9.57 \times 10^8$

Table 1: Top 5 Daily Demand and Revenue by Building Class

Table 1 confirms that trips starting from building class B have the highest daily demand at 517 million trips and the highest daily revenue of $9.39 billion. These top five building classes with the highest demand and revenue are primarily residential areas, with some mixed-use spaces, such as offices or shops[9]. One potential reason is that these residential areas, especially those with mixed-use spaces, are located near amenities and commercial centers. The presence of these building classes in zones mentioned in the previous paragraph may because they have originally been residential areas but evolved into high-demand areas due to their proximity to commercial and cultural centers. High-demand and high-income areas, such as Times Square/Theater District and Midtown Center, which are renowned for business and cultural activities, significantly contribute to the increased demand and revenue.

4 Modelling

To identify the vital factors affecting hourly demand and revenue, four models were developed: Linear Regression[10] (LR) to predict hourly demand and revenue, Random Forest[11] Regressor (RFR) to forcast hourly demand and revenue. The top five features based on feature importance from each of model were selected to determine the critical elements. The input attributes for predicting hourly demand include all features except `log_avg_hourly_demand`, and for predicting hourly revenue, all features except `log_avg_total_revenue` were used.

For the LR models, the following assumptions were made: a linear relationship among the input variables, no interaction between features, no multicollinearity among the independent variables, and independence of errors among observations.

Before training each model, a `VectorAssembler` instance was created to combine the input columns into a single feature vector for both training (July 2023 to November 2023) and testing data (December 2023). This step simplifies the input for machine learning models which ensures efficient model building and tuning. During training process, LR models employed cross-validated grid search[10] for hyperparameter tuning to minimize Root Mean Squared Error (RMSE) and then improve accuracy. RFR required manual tuning due to excessive computation time caused by large dataset size, but successfully minimized RMSE.

5 Results and Discussion

The performance of these models was evaluated using the RMSE, which quantifies the average magnitude of the forecast error and has strong interpretability. From table2, it can be seen that the RMSE for LR is lower in both cases, indicating that it performs better in terms of forecast accuracy and has

RMSE	Hourly Demand	Hourly Revenue
LR	0.437	0.0758
RFR	0.495	0.0846

Table 2: RMSEs for Predicting Hourly Demand and Revenue Using LR and RFR Respectively

a smaller error compared to RFR. These results demonstrate that the LR model generally provides more accurate predictions, indicating that it better captures the underlying patterns in the data. The relatively close RMSE values between the two models indicate that neither of them overfits the data.

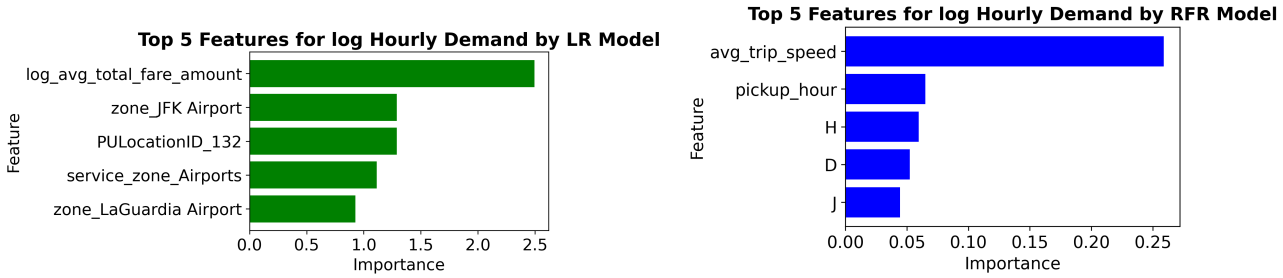


Figure 4: Top Feature Importance of Hourly Demand by LR and RFR

Figure 4 highlights the key attributes that influence the hourly demand determined by the LR and RFR models. The LR model emphasizes high-traffic areas such as JFK (`PULocationID_132`) and LaGuardia Airports, showing that busy regions drive growing demand. Since market positioning to attract specific customer segments by raising prices[12], higher total fare amounts are associated with increased demand. In contrast, the RFR model considers building class and travel efficiency. Specifically, faster travel speeds enable more orders to be completed in the same time, thereby can increase the hourly income. And peak hours within each day such as 8 AM will also boost transportation needs then further increase hourly earnings. Hotels (`H`) generate high demand probably because travelers often lack their own transportation and prefer alternatives. Moreover, hotels and elevator apartments (`D`) are often located in densely populated areas, resulting in a steady flow of passengers needing frequent short trips. Theaters (`J`) further contribute to hourly revenue by creating short-term surges in demand, especially before and after performances.

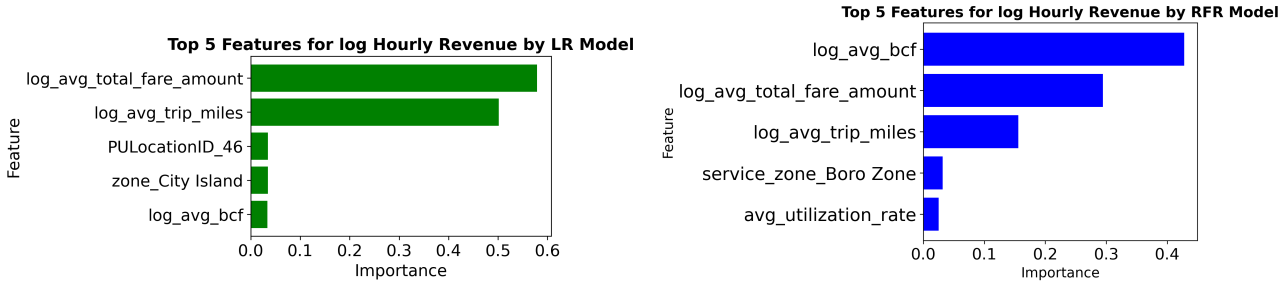


Figure 5: Top Feature Importance of Hourly Revenue by LR and RFR

Figure 5 reports the primary elements affecting hourly revenue selected by LR and RFR models. The LR model emphasizes total fare amount and trip distance, as their higher values lead to increased income directly. On the other hand, the RFR model incorporates other aspects including black car funds, utilization rate, and division of borough zones (`service_zone_Boro Zone`). The black car funds reduce operating costs (such as medical bills if injured on the job and lost wages) and provide various medical insurances[13], thereby improving overall earnings. High utilization rates indicate that drivers

spend less time waiting for dispatch and less time traveling to the destination of orders, thus potentially increasing hourly revenue. In addition, different borough zones can impact earnings due to variations in needs, billing structure, and passenger purchasing power. For instance, high-demand areas such as Manhattan typically have higher ridership and spending power, thus boosting hourly revenue.

6 Recommendations

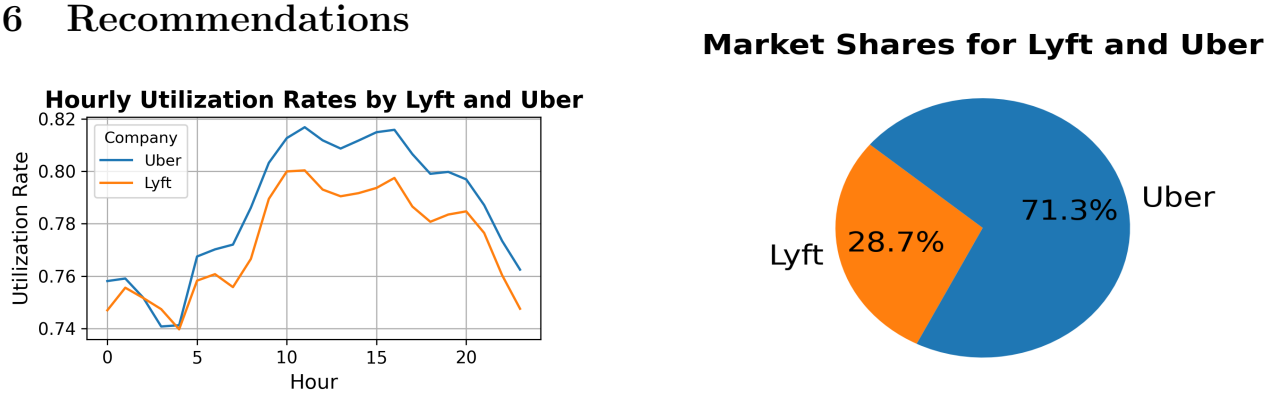


Figure 6: Hourly Utilization Rates and Market Shares by Lyft and Uber

6.1 Targeted Dispatch Optimization for Maximizing Demand and Revenue

Recommendation: Taxi companies should optimize vehicle dispatch by aligning driver availability with peak demand and revenue hours (8 AM and 5-7 PM) mentioned in 3.1 and strategically positioning vehicles in high-demand locations and building classes mentioned in 3.2. This combined approach ensures that drivers are on the road during the most lucrative times of the day and in areas where passenger pickups are most likely, thereby maximizing both hourly demand and revenue.

Justification: Analysis in section 3 and 5 has proved that specific hours of the day have significantly higher trip volumes and earnings. Additionally, geospatial analysis identifies certain areas, such as busy commercial districts, residential regions, and entertainment zones, that consistently experience high passenger needs. Notably, the highest utilization rates for Lyft and Uber, as shown in the left diagram of Figure 6, occur at 11 AM and 4 PM, which do not align with the peak hours of hourly demand and revenue. Therefore, by strategically deploying more drivers during these peak hours and targeting the high-demand locations, taxi companies can optimize the vehicle dispatch strategies, increase utilization rates, reduce idle time, and enhance overall efficiency. Furthermore, this data-driven planning could also improves driver earnings by ensuring they are available when and where they are most needed.

6.2 Boost Driver Revenue by Utilizing Uber's Market Share and Black Car Fund

Recommendation: Taxi companies should encourage drivers to make use of the benefits of Black Car Fund and leverage Uber's market dominance to optimize revenue. In other words, by ensuring drivers receive health, safety and financial support from the Black Car Fund, companies can improve driver satisfaction and performance, resulting in attracting more consumers and increasing revenue.

Justification: Figure 5 affirms the importance of black car fund in revenue. Moreover, as shown in the right diagram of Figure 6, the market share of Uber far surpasses that of Lyft. This significant market presence and high utilization rate create substantial earning opportunities for drivers by tapping into large customer base of Uber. By taking advantage of the benefits offered by the Black Car Fund, such as accident disability insurance, dental insurance, and psychology and safety support[13], drivers can greatly reduce their personal expenses while gaining financial stability. This support enhances job security and motivation, leading to improved service quality. Consequently, higher driver satisfaction and loyalty might attract more passengers, boosting revenue for both the company and its drivers.

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